PICO 2 / PICO9918 BENCH PROTOTYPE

Direct bus + SWD | 40-pin ZIF | Manual power | 12 September 2026

3.3 V data writes: confirm target U5 is HCT245. Saved schematics say HC245; direct writes then need bench validation.
One USB source powers both boards through Pico 2 VBUS. GP26-GP28 are NOT 5 V tolerant; no bus wires use them.

PROBE: PICO 2 HEADER

TARGET: ZIF SOCKET

GPIO	Physical pin		DIP pin	Target header
GP4	6	← **D0 / CD7** →	**17**	H2.7
GP5	7	← **D1 / CD6** →	**18**	H2.8
GP6	9	← **D2 / CD5** →	**19**	H2.9
GP7	10	← **D3 / CD4** →	**20**	H2.10
GP8	11	← **D4 / CD3** →	**21**	H3.1
GP9	12	← **D5 / CD2** →	**22**	H3.2
GP10	14	← **D6 / CD1** →	**23**	H3.3
GP11	15	← **D7 / CD0** →	**24**	H3.4
GP12	16	— **/CSR** →	**15**	H2.5
GP13	17	— **/CSW** →	**14**	H2.4
GP14	19	— **MODE** →	**13**	H2.3
GP15	20	— **MODE1** →	**11**	H2.1
GP21	27	— **/RST** →	**34**	H4.2
GP18	24	← **/INT** —	**16**	H2.6
GP19	25	← **CPUCLK** —	**38**	H4.6
GP20	26	← **GROMCLK** —	**37**	H4.5
GND	23	— **GND** —	**12**	H2.2

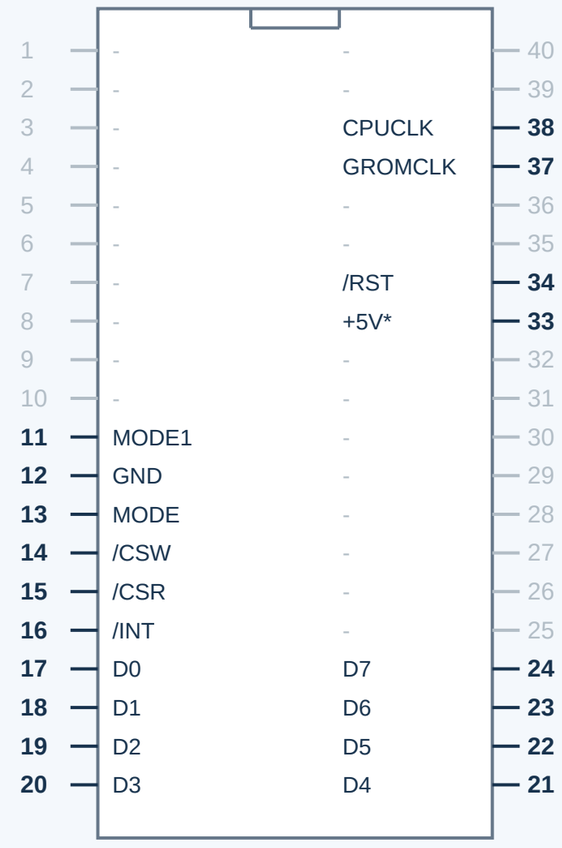
Arrows show signal direction; data pins switch between input and output in firmware.

SWD: SEPARATE TARGET DEBUG CONNECTOR

GP2 / pin 4	— **SWCLK** →	DBG1 pin 1
GND / pin 3	— **GND** —	DBG1 pin 2
GP3 / pin 5	← **SWDIO** →	DBG1 pin 3

ZIF: TOP / INSERTION VIEW

Align target pin 1 with socket pin 1.



* Pin 33: Pico VBUS via jumper or page-2 switch.

15.24 mm target rows; socket body not to scale.

POWER + VIDEO

PC USB -> Pico 2 USB; VBUS pin 40 -> removable jumper -> ZIF pin 33. Leave target USB unplugged.
Target video FFC -> existing VGA adapter -> monitor. No video signals go through the probe.

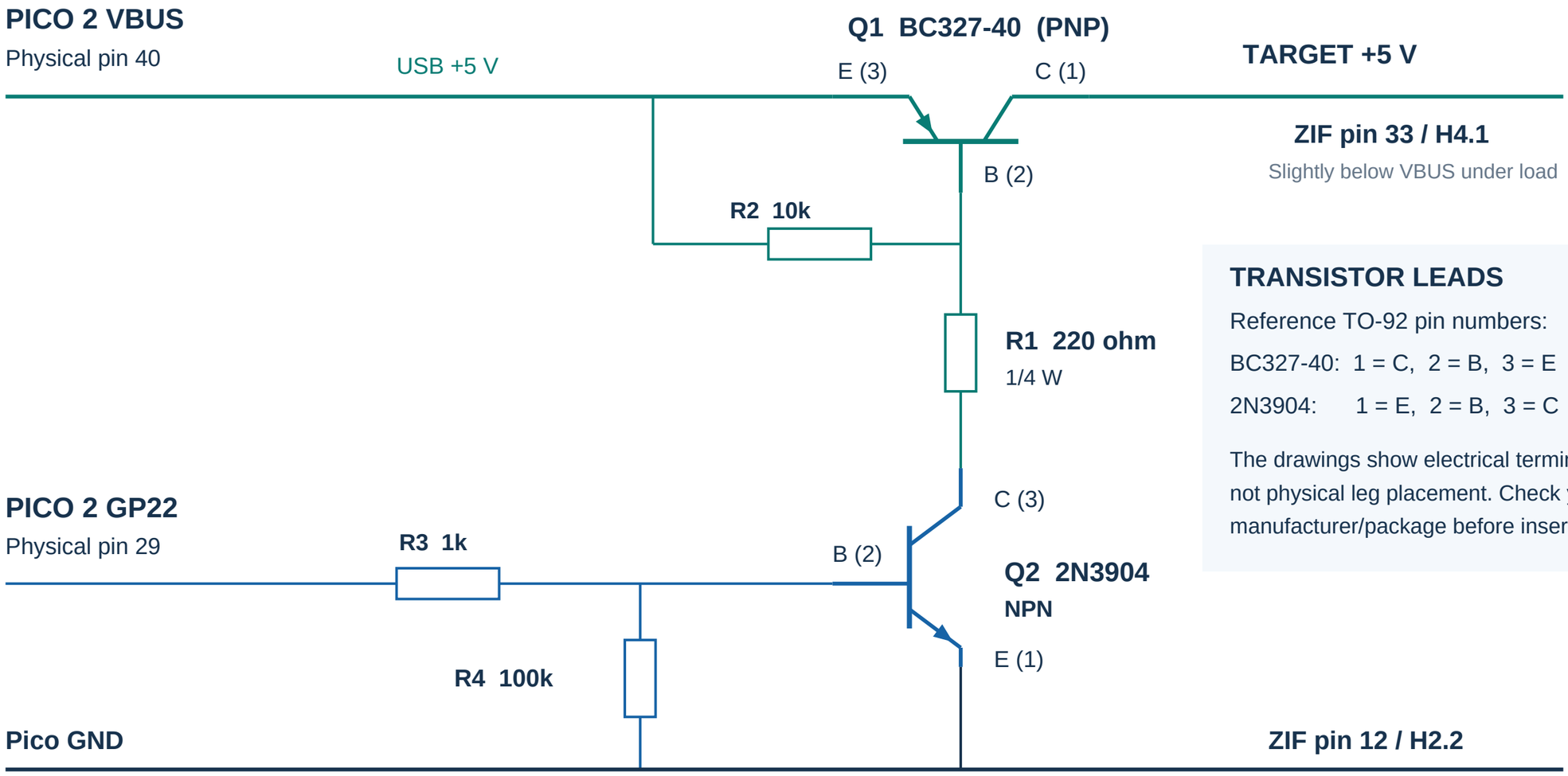
1. Configure /CSR, /CSW, MODE, MODE1 and /RST idle HIGH, then fit the target power jumper.
2. Make GP4-GP11 inputs BEFORE asserting /CSR. Release /CSR and allow bus turnaround before driving data.
3. For a manual power cycle, release bus/SWD outputs and pulls, remove the power jumper, then reconnect.

Manual option shown. Optional transistor power switch: see page 2. Use GP2/GP3, not the probe native SWD pads.

OPTIONAL TARGET POWER / PARTS ALREADY ON HAND

BC327-40 + 2N3904 + four resistors | GP22 HIGH = power ON | LOW / floating = OFF

Replace the page-1 power jumper with this switch. Leave target USB unplugged; retain all signal/ground wires.
Q1 switches +5 V. Ground remains permanently connected. No target 3.3 V connection.



TRANSISTOR LEADS

Reference TO-92 pin numbers:

BC327-40: 1 = C, 2 = B, 3 = E

2N3904: 1 = E, 2 = B, 3 = C

The drawings show electrical terminals, not physical leg placement. Check your manufacturer/package before inserting.

Common ground: same as page 1

HOW IT WORKS

GP22 drives Q2, which pulls Q1 base down through R1. Q1 then supplies the target. R2 and R4 default both transistors off.
R1 gives about 18 mA base drive to Q1; R3 draws about 2.5 mA from the Pico. Initial design load: up to 150 mA target current.

Check the loaded voltage at ZIF pin 33 before relying on this switch: aim for at least 4.5 V with the target running.

Before GP22 LOW, release bus/SWD outputs and disable pulls. Start with a 1-second off delay; verify that the target resets.

Pico USB must supply probe + target + base drive; request a 500 mA USB configuration before enabling target power.

Prototype switch: no current limiting or active discharge. No relay or flyback diode is needed. See README for connections.